

B894: Dissertation

Artificial Superintelligence & Applicant Attraction: Can ASI Substantially Influence Neurodivergent Employee Experiences?

Submitted in partial fulfilment of the MSc in HRM at The Open University

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Contents

1 Acknowledgements	3
1.1 Outline of the Topic.....	3
2 Introduction	4
3 Research Methodology	4
4 Why this topic is important.....	4
5 Timeline and Next Steps.....	5
6 Introduction to TMA 2 – Literature Review.....	5
7 Mapping subtopics that are relevant.....	6
7.1 Conducting Gap Analysis – ASI and HRM	6
8 Explicating my Literature Review 1 – Applicant Attraction	7
9 Explicating my Literature Review 2 – Signalling Theory.....	8
10 Explicating my Literature Review 3 – Recruitment & Selection.....	9
11 Journal of Economic Literature (JEL) Classification.....	9

1 Acknowledgements

I am forever grateful to Almighty God my strength, my fortress, my redeemer. My parents for being supportive of this endeavour I have been called to pursue and my course mates for endless support along the way.

Throughout the experience of postgraduate studies at The Open University, I have been supported by several internal departments, external organisations and numerous people in academia who have worked endlessly to craft my critical understanding of Human Resources and who have helped define my research topic around employee wellbeing. These include Library Research Support, Student Support and of course my former lecturers and current tutors.

I would lastly like to take this opportunity to convey my gratitude and a tremendous thank you to all of those who have helped finance my learning and research endeavours, either through tuition loans or grants, particularly I would like to draw attention to the efforts of Student Finance England (SFE) and the Open University Student Budget Account (OUSBA) who have helped me settle in and funded my pursuit throughout my time as a postgraduate student.

The TMA 1 assignment began here

1.1 Outline of the Topic

This opportunity to prepare a MSc dissertation thesis all emerged because of the Buckland Review which was commissioned by Sir Robert Buckland QC (the long-serving former UK Cabinet Minister and former MP for South Swindon) and Autistica in early 2024 under the leadership of the then Conservative Prime Minister, the Rt. Hon Rishi Sunak.

Since the outcome of the Buckland Review and its subsequent findings concerning the employment opportunities of Autistic applicants (that is, 7 in 10 are not in gainful employment across the country), further research has been conducted by Miller and Doyle (2025) which forms the basis of how to *deconstruct* the various approaches taken by employers when attracting these aforementioned and completely hypothetical 7 in 10 neurodiverse applicants who are unemployed during recruitment activities, without employing specific and targeted recruitment campaigns which require candidates to come forward and identify their conditions at the outset of the process. Dr. Nancy Doyle is a prolific campaigner for neurodivergence in the workplace who has previously written an online opinion piece for Genius Within, a UK community interest company, on the subject of the Sir Charlie Mayfield Review (Doyle, 2025).

The aforementioned is important for several reasons, not least for the intrinsic motivation of the employee but also on grounds of fairness and positive discrimination. I am aiming to study how neurodiverse applicants perceive resource-based HRM (what we call the RBV in HRM theory) and how RBV approaches, which are inherently focused on creating competitive advantage link to recruitment signalling during the applicant journey.

I am hoping to investigate several hypotheses (which I have cited in my PhD proposal) which will form the backbone of the research study, and which will either confirm the work of Miller and Doyle (2025) or reject the premise of their recent research study. I also plan to extend Miller and Doyle's work to study anonymous responses from applicants with bipolar disorder (a condition that is widely considered as neurodivergent but was omitted from their empirical study).

2 Introduction

Given that recent research demonstrates how neuroinclusive teams can be 30% more productive than those which do not include neurodiverse employees (NatWest Mentor, 2025), and with Gen Z entering the workforce – over 50% of whom have some sort of neurodivergent condition² - either Autism Spectrum Disorder or Bipolar Disorder (Forbes Magazine, 2025), the broad research topic I have selected to follow for this module is a contemporary problem with its roots in a recent report (Youth Future's Foundation, 2025). As an interdisciplinary topic, this thesis is partly covered by *signalling* (psychology), *applicant attraction* (HRM), *employee wellbeing* (HRM) and *labour market analysis* (game theory).

As I am deeply intrigued by the sort of opportunities available to young and neurodiverse job applicants and how these opportunities are perceived by those who apply, I have chosen a topic that investigates an instance and prevalent gap in the literature on hiring technology and perceptions of website recruitment processes by neurodiverse applicants (Miller and Doyle, 2025). I have devised a set of five hypothesis to outline the direction which I would like the research to follow, and which resulted in the research question I have settled on.

3 Research Methodology

For this Masters' thesis, I will be aiming to build on the MSc in HRM research study (which applies a sentiment analysis and a questionnaire approach) by conducting empirical analysis of a diverse range of data sets including non-sensitive HR data from companies. The research questionnaire is available (in draft) to view [here](#) and resumes the research topic where (Miller and Doyle, 2025) ended, although the topic of discussion has been realigned to include artificial intelligence, the fundamental assertions are the same. The following hypotheses are subject to supervisory changes and will be honed and amended as and when new information in the literature is discovered, reviewed and internalised. In order, these initial hypotheses read as follows:

Hypothesis 1: Firms with resource-based HRM practices signal *more* in successful applicant "journeys"

Hypothesis 2: Firms who signal more, also spend more annually on recruitment processes

Hypothesis 3: Firms with higher net spend "bundles" enable rather than deter neurodiverse applicants

Hypothesis 4: Neurodivergent applicants judge recruitment favourably compared to other colleagues

Hypothesis 5: Firms having lower net spend signal less resource-based HRM practices

These hypotheses refer to the research designed by Chang and Chin (2018) in their paper on online ratings and signalling and act as a starting point for further hypotheses. The theories which underpin this interdisciplinary topic can be traced back from its applied influences in HRM to what Guest (2021) refers to as attribution theory – the theory of psychology which explains how behaviour is understood as a consequence of one's environment. I will aim to investigate how signalling (Spence, 1973) affects situational attribution theory, and how it is thereby relevant to applicant attraction of neurodiverse candidates (Celani and Singh, 2011). I have since applied for the PhD at the Open University and the research question I have submitted can be drafted as:

"Artificial Intelligence & Applicant Attraction: Can AI Substantially Influence Neurodivergent Candidate Experiences? *An empirical analysis*"

4 Why this topic is important

The importance of this research topic cannot be understated. Specifically, the topic is important because it seeks to enquire into how diverse job applicants perceive their roles in the organization *prior to* being onboarded as employees (in other words, it asks how *neurodiverse* applicants interact as job candidates)

² I define two specific conditions in-scope of this study: Autism and Bipolar Disorder.

through the proxy of website recruiting activities. The seminal applied instance of this research topic is the online ratings and HRM signalling paper pioneered by Chang and Chin (2018) who test evidence to investigate the hypothesis that three things are likely to occur prior to onboarding:

1. A high-commitment HRM strategy is pursued and that firms conduct recruitment via a website
2. The firm may initiate a signal to the receiver via this website about its high-commitment HRM strategy
3. This signal initiates an effect on the employees' online ratings.

As such, the authors (Chang and Chin, 2018) argue for five important hypotheses which form the starting point for this study:

- 1) Firms with more high-commitment HRM will signal more HRM information on websites
- 2) Firms signalling more HRM information on websites will experience a trend of more applications
- 3) Firms' signalling of HRM information on websites fully mediates the effect of the firms' commitment HRM on recruitment
- 4) Firms' *actual* HRM influences employees after experience outcomes such that firms' commitment HRM will increase online ratings
- 5) Firms' signalling of HRM will not increase an employees' online ratings if the effect of commitment HRM (on recruitment) is controlled

My research will aim to extend a gap on this topic by focusing on firm expenditure, capabilities and resources, asking a question not currently addressed by the literature on signalling – how these factors affect competitive advantage given the specific context of attribution theory (Guest, 2021). As such, there are at least three forms of data will be required to accomplish the task of addressing the question:

1. Data on HRM signals from recruitment websites and publicly available announcements
2. Data drawn from surveys on how neurodiverse applicants perceive signals on recruitment websites
3. Data on how firms spend on recruitment websites

5 Timeline and Next Steps

Out of scope: I will be conducting a literature review for B894 which will be completed in February 2026, I do not intend to look at any sort of data or make any requests for data access until ethical approval has been given for my MSc research question and methodological design.

Within Scope: I will begin to map the hypotheses using sources such as Overton.io and others that have been identified; and assuming the research question can be scrutinised on an ongoing basis, it will then be possible to review the available literature, filter through and arrange such information into formats which will allow me to revise the hypotheses (cited above), and hopefully make use of both quantitative methods such as sentiment analysis and indeed qualitative analysis such as a survey or questionnaire.

The TMA 2 assignment begins here

6 Introduction to TMA 2 – Literature Review

As SHRM (specifically, RBV) is a very interdisciplinary subject when it comes to research that defines theory - I will be sourcing information on the following subtopics from three highly reliable contemporary literature sources (JSTOR, The Open University Library and Google Scholar). My intention is to source academic peer-reviewed literature published roughly between 2010 – 2025 in terms of timelines (the literature effectively narrates on the problem which can be traced back to about 2011 in reports so, there's about a 8–10-year gap where the problem surfaces and the seminal papers focused on analysing the problem begin to emerge in the literature). However, there may be exceptions to this timeline as per the theoretical work of Coles and

Treble (1993) or the earlier theoretical work by Rynes and Barber (1990). Though the three selected sources indeed do store great literature on each subtopic, I may require policy infographics, charts and vital published statistics which will also be important to this study's objective. I will therefore be actively seeking NGO and government reports drawn from Overton.io, where such information may be found.

7 Mapping subtopics that are relevant

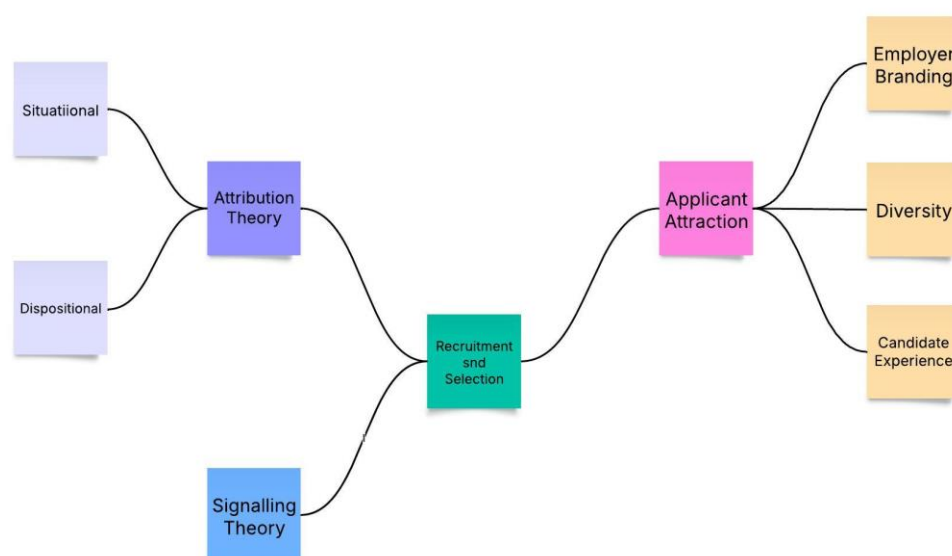
Recruitment and Selection: According to Sir Robert Buckland's Autism and Employment Review published alongside Autistica on the employment of sufferers on the ASD spectrum, only 3 in 10 people with ASD are gainfully employed, the remaining 7 in 10 of the UK's population are unemployed or underemployed. This figure rises to 8 in 10 on a worldwide basis (Davies et. al. 2023).

Applicant Attraction: A recent study on applicant attraction by Dutta and Mishra (2021) have focused on Gen-Z and millennial generation cohorts in the context of the Indian sub-continent.

Signalling Theory: The seminal work on signalling is found in Spence (1973) however, since this paper was published there have been numerous attempts to assign and apply signalling theory to the context of the hiring process.

Attribution Theory:

Figure 1: Mind Map of Relevant Literature Subtopics (Stage 2 Project Status)



7.1 Conducting Gap Analysis – ASI and HRM

While some gaps certainly do exist in the form of “*minimal information*” available concerning the effects of bipolar depression in later life (Depp and Jeste, 2004), one of the more pronounced gaps in the literature is the more recent focus on Artificial Intelligence and employment balances for neurodivergent neurotypical applicants (Mirowska and Arsenyan, 2025; Smith *et. al.*, 2025) which assumes the capture of those who suffer from Bipolar I or II. Artificial Superintelligence or “ASI” has been argued to carry the potential to promote or affect various types of diversity, to avoid the consequence of overexposure to unconscious biases, and to help design more comprehensive signalling approaches to talent attraction, recruitment and an organisational focus on *cultural* diversity more generally. According to an open access paper by Chilunjika,

et. al. (2022), gaps exist in the research on AI's application to HRM in the public realm. Chilunjika, *et. al.* (2022) refers explicitly to a separate paper by Jia, Gao, Li and Li (2018).

- AI as an umbrella term for concepts machine learning, NLP, computer vision and other key algorithmic concepts

8 Explicating my Literature Review 1 – Applicant Attraction

As part of the planning process, I shall spend time gathering articles from a number of relevant online databases which host peer-reviewed journals. These are listed below under the "Source" column. I plan to gather a minimum of five articles from each of the clear *subtopics*, acting as search protocol and using the jigsaw format as structure (University of Hull Library, 2025).

1. Applicant Attraction

Source	Name of Journal / Book	Scoped Articles
Emerald	Personnel Review	Celani and Singh (2011)
JSTOR	Academy of Management Review	Rynes and Barber (1990)

2. Neurodiversity (Bipolar and Autism - Diversity and Inclusion)

Source	Name of Journal / Book	Scoped Articles
Wiley Online	The Scandinavian Journal of Economics	Sarsons (2024)
Emerald	Personnel Review	Dutta & Adsule (2025)
Wiley Online	Human Resource Management	Lup and Canonico (2024)
Sage Journals	Autism	Davies et. al. (2023)

3. Recruitment and Selection (Game Theory and Signalling Theory)

Source	Name of Journal / Book	Scoped Articles
Emerald	Personnel Review	Younis & Hammad (2021)
Wiley Online	The Scandinavian Journal of Economics	Barmby, Sessions & Treble (1994)
Elsevier	Labour Economics	Coles & Treble (1993)
Elsevier	Economic Letters	Coles & Treble (1996)

Emerald	Personnel Review	Celani & Singh (2011)
Wiley Online	Journal of Occupational & Org Psychology	Miller & Doyle (2025)
Wiley Online	Human Resource Management Journal	Guest et. al. (2021)
Elsevier	Journal of Business Research	Chang & Chin (2018)
JSTOR	The Academy of Management Review	Bowen & Ostroff (2004)
JSTOR	The Quarterly Journal of Economics	Spence (1973)
Wisconsin.edu	Master of Science (Psychology) Dissertation	Wright (2010)
EBSCO Host	Academy of Management Journal	Jones et. al. (2014)

9 Explicating my Literature Review 2 – Signalling Theory

Key question: What is the sub-game optimal (in a game of separating or pooling equilibrium) signal an employer should send to applicants in order to be neuroinclusive? Also, at what level of signal does an employer who signals acceptance of a neurodiverse condition, become accepted by a neurodiverse applicant?

1. Spence (1973): Theoretical Article

- Here, hiring is compared to the act of buying a lottery ticket
- As per Brown and Sessions (2004), there is often an element of informational asymmetry; full knowledge of the workers capability in the job prior to lottery is rarely fully disseminated
- An employer's perception of this lottery gamble is what Spence defines as his "primary interest" due to the uncertainty involved and the wages offered by the employer
- There are two kinds of observable attributes mentioned in the article:
- Indices and signals (p. 357)
- The term "indices" refers to "immutable" attributes, such as race or sex, and "signals" references alterable attributes such as education and experience. Age as an alterable attribute is considered to be an index
- Signals and indices are seen to be "parameters" in the conditional probability distribution reflecting the "beliefs" of an employer

2. Miller and Doyle (2025): Empirical Article

- Mentions the theoretical framework of Khan et. al. (2023) whose literature review and empirical study on the topic is the only recent work used as authority, but acknowledges research in the area of inclusive signalling to neurodiverse applicants is relatively limited
- Extends Khan et. al. (2023)'s ideas for further research in a HR context and applies a systems-based analysis of "signalling" theory (or rather a systematic approach based on academic literature) in the hiring and selection process to cover the inclusion of the neurodiverse job applicant in the workplace
- Argues organisations should go *beyond* the business case for diversity in recruitment. But how do Miller and Doyle (2025) suppose this should be achieved?

3. Celani and Singh (2011): Theoretical Article

- An analysis of signalling theory from a multi-level perspective (contextual effects)
- Recruitment-related signals are seen to reveal unknown characteristics of organizations
- Theory links between individual and organisation-level signals from recruitment activities
- Links are made between market signals and instrumental or symbolic applicant inferences

4. Chang and Chin (2018) Empirical Article

- The authors define signalling as the communication of unique information or rather "the sending out of information that competitors cannot easily imitate"
- Signalling becomes important in job markets where numerous employers compete to "increase market share"
- Importance of firm actions and separating equilibrium to this signalling dynamic
- The authors analyse HRM signalling on each firm's website over a three-year period

5. Guest et.al. (2021) Empirical Article

- Studies staff in bank branches
- Defines signalling theory as “communication by and within an organisation”
 - In contrast to evidence from Laker et. al. (2025)
- Cites Bowen and Ostroff (2004) as part of an integrated framework to study attribution theory
- Concludes that signalling theory is to be viewed as a “*distinctive*” approach to the HRM process
- Acknowledges limitations of cross-sectional methodology and calls for longitudinal approaches
 - Bowen and Ostroff (2004)
 - Touches on multi-level relationships between HRM and organisational performance through psychological “climate”³ and organisational sensemaking (a design thinking principle). (p.205)
 - Highlights the additional importance of “social context theory”
 - Formal roles are defined by organisational systems and “maintenance subsystems”
 - Distinctiveness: “Visibility, Understandability, Legitimacy of authority and Relevance”

10 Explicating my Literature Review 3 – Recruitment & Selection

Key question: In a separating equilibrium, that is, in a sub-game perfect Bayesian repeated game with known payoffs, and standard assumptions (complete information and perfect monitoring), how do employers navigate a many-to-one “lottery” scenario, using the offered wage as signals as navigating mechanisms?

1. Coles & Treble (1993; 1996) are some of the key seminal works on the dynamics faced by the employer and the behaviour of the employer when faced with uncertainty.
2. TBC

11 Journal of Economic Literature (JEL) Classification

To navigate the literature, I shall be using several JEL codes to classify my MSc in HRM dissertation, these codes are:

JEL CODE	PRIMARY/SECONDARY CATEGORIES
J24	Human Capital; Skills; Occupational Choice; Labor Productivity
J71	Labor Discrimination
D82	Asymmetric and Private Information; Signalling
D83	Search; Learning; Information and Knowledge; Communication; Beliefs
D86	Economics of Contract: Theory
D90	Micro-Based Behavioural Economics
M51	Personnel Economics; Executive Compensation
M54	Labor Management; Industrial Relations
I12	Health Behaviour; Health Economics
I14	Health and Inequality

Source: OpenAI (2025)

³ As an example of higher-order social structures (other examples incl. organisational culture and organisational role structure).

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